

5.1 Machine Learning, Neural Networks & Deep Learning

مصنوعی ذہانت (AI) سب سے بڑا میدان ہے۔

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 01 of 15

5.1 Machine Learning, Neural Networks & Deep Learning

LEARN IT

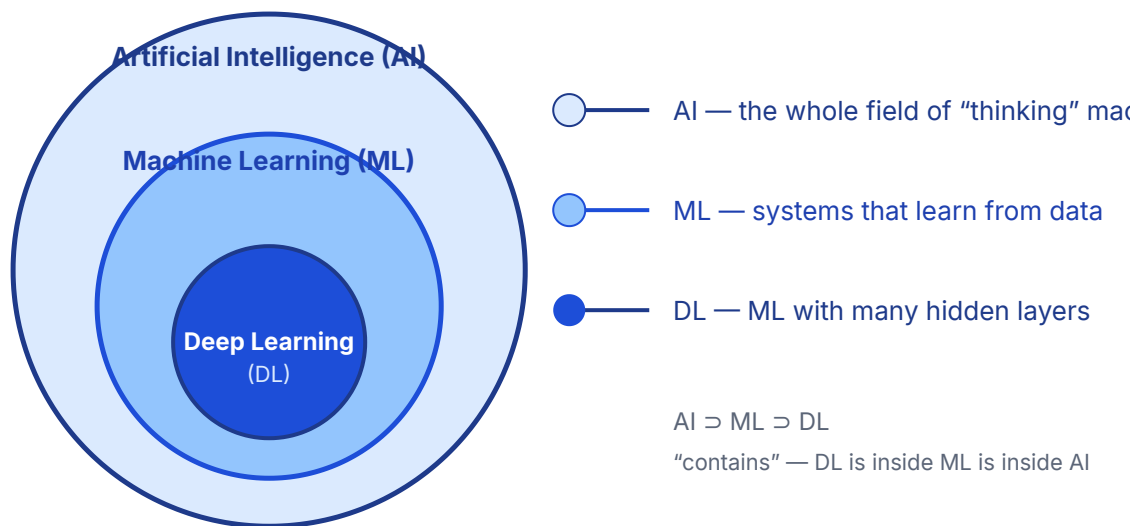
Machine learning (ML), **neural networks (NN)** and **deep learning (DL)** are important areas of **artificial intelligence (AI)**. **AI** is the broad field in which computers perform tasks that normally need human thinking, such as recognising a face, understanding speech or making a decision.

These technologies help computers **learn from data** and make useful decisions **without being explicitly programmed for every single step**. You do not write a separate rule for every possible face or every Urdu accent. You show the computer many examples, and it finds the patterns itself.

They are used in many modern systems:

- **Face unlock** on smartphones.
- **Speech recognition** (for example Siri, Alexa, or Urdu voice typing on a phone).
- **Photo search** categorisation (typing "car" in a gallery and seeing car photos).
- **Medical diagnosis** from X-rays and scans.
- **Language translation** (English to Urdu or Sindhi).

Remember the nesting: **AI is the biggest set**. Inside AI sits **ML**. Inside ML sits **DL**. Every deep-learning system is a kind of machine learning, and every machine-learning system is a kind of AI — but not every AI system is deep learning.



The set relationship $AI \supset ML \supset DL$. Deep learning is a method of machine learning; machine learning is a part of artificial intelligence.

WORKED EXAMPLE

Worked example: place each system in the correct set. Sara in Karachi uses three features on her phone.

1. **A chess app that follows fixed if-then rules** written by a programmer (“if the queen is attacked, move it”). This is **AI** (the machine decides), but it is **not ML**, because it does not learn from data.
2. **A JazzCash fraud filter** that was trained on thousands of past transactions and then flags a new unusual payment. This is **ML** (and therefore also AI). If it uses only a simple model with few layers, it may **not** be DL.
3. **Face unlock** that studied millions of face images through many hidden layers. This is **DL** — and therefore also ML and AI.

Exam sentence you can write: *All DL is ML, all ML is AI, but not all AI is ML, and not all ML is DL.*

REMEMBER IT

Russian-doll memory trick. Picture three boxes. The outer box is **AI**. Inside it is a smaller box **ML**. Inside that is the smallest box **DL**. You cannot take DL out of ML, and you cannot take ML out of AI.

COMMON MISTAKE

Mistake: “AI, ML and DL are three different, separate subjects.”

Fix: They nest. $DL \subset ML \subset AI$. A deep-learning face-unlock system is all three at once.

مصنوعی ذہانت (AI) سب سے بڑا میدان ہے۔ اس کے اندر مشین لرننگ (ML) ہے، اور مشین لرننگ کے اندر ڈیپ لرننگ (DL) ہے۔ یہ ٹیکنالوجیز کمپیوٹر کو ڈیٹا سے سکھاتی ہیں تاکہ ہر قدم الگ الگ پروگرام کیے بغیر فیصلے کیے جا سکیں، جیسے چہرے سے فون کھولنا یا آواز پہچاننا۔

CHECK YOURSELF

1. Which is the largest set: AI, ML or DL?
2. Is every neural-network face-unlock system an example of AI? Why?
3. Give one modern system from this section that uses these technologies.

Answers: 1. AI (AI $\supset$ ML $\supset$ DL). 2. Yes — DL is inside ML, and ML is inside AI. 3. Any of: face unlock, speech recognition, photo search, medical diagnosis, language translation.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. ML, NN and DL are parts of:

- a. AI
- b. OSI layer 1 only
- c. Access wizards only
- d. Waterfall only

MCQ 2. They all need:

- a. Data to learn from
- b. Only copper cable
- c. Only primary keys
- d. Only Figma

MCQ 3. The course order is:

- a. ML as the idea, NN as a model, DL as deep NNs
- b. DL then stone tools
- c. SQL then NAND
- d. HCI then K-maps

MCQ 4. Pakistan's AI policy is mentioned as:

- a. A future/local context
- b. A Python library
- c. A chart type
- d. A file mode

MCQ 5. A student mistake is:

- a. Treating the three terms as identical
- b. Giving one example
- c. Mentioning data
- d. Mentioning layers

MCQ 6. This lecture is the map; later ones:

- a. Define each term properly
- b. Delete the chapter
- c. Only careers
- d. Only MVPs

Extra short questions

1. Place ML, NN, DL in one nested sentence.
2. What do they share?
3. What should an XI/XII answer still include?
4. One Pakistani hope from the notes.

Extra long question

1. Draw a nested diagram $AI \supset ML \supset NN \supset DL$ and write one example at each ring.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. DL is a kind of NN which is a way to do ML which is a kind of AI.

2. Learn patterns from data instead of only hand-written rules.
3. A human-responsible, data-quality, bias warning.
4. Agriculture, health, Urdu NLP, public service — any from the book.

Long

1. AI: chess rules engine. ML: spam filter. NN: digit recognition. DL: speech-to-Urdu. One limit at the DL ring (needs lots of data/compute).

Next lecture: 5.1.1 Machine Learning (ML)